Hemadri Shekhar Das

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Education

Chennai Mathematical Institute Chennai, India
MSc. CS Student - 7.94 cgpa *Aug. 2024 - July 2026*

Favourite courses attended:

- Randomized Algorithms
- Networking Fundamentals
- Advanced Networking
- Introduction to concurrent programming
- Advanced Concurrent programming
- Cryptographic Standards and Cryptanalysis

My Master's Thesis was on trying to improve upon the CONGA load balancer in data center. To do that I implemented CONGA in ns3, compared it to standard ECMP to make sure our results are in line with the CONGA paper. After that we tried to combine power of 2 choices with CONGA to improve CONGA. At the end we propose a new algorithm that could perform on par with CONGA but without the need for custom ASICS.

Asutosh College, Calcutta University Kolkata, India
BSc. Mathematics Honors - 7.53 cgpa *Aug. 2020 - 2024*

Notable Courses Attended:

- Real Analysis
- Group Theory - I & II
- Ring Theory
- Multivariate Calculus - I & II
- Probability and Statistics
- Linear Algebra - I & II
- Linear Programming and Game Theory
- Numerical Methods

The Modern Academy High School Kolkata, India
ISC - 92.5 % *2020*

Ram Mohan Mission High School Kolkata, India
ICSCE - 95% *2018*

Projects

RTT-Aware Flowlet Routing for Datacenter Networks

Research Project | Extension of MSc Thesis, Chennai Mathematical Institute *May 2025 - Oct. 2026*

- Investigated whether **approximate RTT measurements could serve as an implicit congestion signal** for adaptive datacenter load balancing, reducing reliance on explicit fabric congestion measurements and feedback mechanisms such as CONGA.
- Designed and implemented an **RTT-aware flowlet routing protocol in ns-3.48**, using packet-tag-based RTT feedback, EWMA smoothing, probabilistic path selection, and fixed/adaptive flowlet timeouts.

- Developed multiple path-selection strategies, including **RTT-weighted probabilistic routing and Power-of-Two Choices**, and evaluated them against ECMP and CONGA on symmetric and asymmetric leaf–spine networks.
- Conducted controlled experiments using enterprise traffic distributions across varying network loads, analyzing mean/tail flow completion times, packet loss, and delivery correctness; identified configurations where RTT-aware routing approached or exceeded CONGA's mean-FCT performance, alongside limitations in tail latency and practical deployment.

Arduino and Esp-32 Morse Code Generator

Independent Project *Sept. 2023 - Jan. 2024*

- Developed a Morse code signaling system using an ESP32-C3 microcontroller and the Arduino framework. The system operates as a wireless access point, allowing users to connect via a web interface for input and customization, to control an AC 220V light bulb to generate morse code using a sold-state low-level relay. The web interface includes a login system, adjustable transmission speed, and controls for start, stop, pause, and resume functions.
- FreeRTOS was implemented to efficiently manage multitasking, with separate tasks that handle Morse code generation, relay control, and Wi-Fi connectivity. The output system features two stages: test output and main output with a stopwatch displaying elapsed time and highlighting transmitted characters.
- This project was created to assist my father in teaching the Second Mate Officer course at his marine training institute.

Skills And Languages

Programming Languages: C, C++, Java, Python, Javascript

Markup, Styling & Typesetting Languages: HTML,CSS, LaTeX

Language Proficiency:

- Bengali (Native)
- Hindi (Fluent)
- English (Fluent)

Current Interests

- Computer Networking (e.g., datacenter switches, high-performance networking, distributed AI networking)
- Embedded Systems (e.g., Arduino, ESP-32 for networked IoT projects)
- Home Labs (e.g., self-hosting, Docker, Proxmox, NAS for networking setups)